

Packet Tracer – WAN Simulation Using Serial PPP Authentication

Objectives

Part 1: Build a WAN Topology Using Serial Links

Part 2: Configure Router Interfaces and IP Addressing

Part 3: Configure PPP Encapsulation

Part 4: Configure CHAP Authentication

Part 5: Configure Static Routing

Part 6: Verify WAN Connectivity

Background / Scenario

You are the network administrator for two branch offices connected through a WAN serial link. Instead of using Ethernet, the routers communicate over a point-to-point serial connection that simulates a leased line provided by a service provider.

To improve security, the WAN link will use **PPP (Point-to-Point Protocol)** with **CHAP authentication**. Each router must authenticate itself before traffic is allowed to pass across the WAN.

In this lab, you will:

- Configure serial interfaces
- Configure PPP encapsulation
- Implement CHAP authentication
- Configure static routes
- Verify end-to-end connectivity

This lab introduces WAN technologies commonly found in older enterprise and ISP environments while reinforcing routing fundamentals.

Instructions

Part 1: Build the WAN Topology

Step 1: Add the Devices

a. Open **Cisco Packet Tracer**.

b. Add:

- 2 Routers (2911)
- 2 PCs

c. Rename devices:

- Router0 → **R1**
 - Router1 → **R2**
 - PC0 → **PC-A**
 - PC1 → **PC-B**
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Step 2: Install Serial Modules

a. Click **R1**.

b. Go to **Physical** tab.

c. Power off the router.

d. Insert a **HWIC-2T** serial module.

e. Power on the router.

f. Repeat for **R2**.

Step 3: Connect the Devices

Use the following cables:

Device Port		Device Port	Cable
PC-A	FastEthernet0	R1 GigabitEthernet0/0	Copper Straight-Through
PC-B	FastEthernet0	R2 GigabitEthernet0/0	Copper Straight-Through
R1	Serial0/3/0	R2 Serial0/3/0	Serial DCE

The router connected to the DCE end will provide clocking.

Wait for interfaces to initialize.

Part 2: Configure IP Addressing

Step 1: Configure PC-A

Click **PC-A > Desktop > IP Configuration**

Enter:

- IP Address: 192.168.10.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.10.1
-

Step 2: Configure PC-B

Click **PC-B > Desktop > IP Configuration**

Enter:

- IP Address: 192.168.20.10
 - Subnet Mask: 255.255.255.0
 - Default Gateway: 192.168.20.1
-

Step 3: Configure R1 Interfaces

enable

configure terminal

interface g0/0

ip address 192.168.10.1 255.255.255.0

no shutdown

interface s0/3/0

ip address 10.1.1.1 255.255.255.252

encapsulation ppp

clock rate 64000

no shutdown

end

Step 4: Configure R2 Interfaces

enable

configure terminal

interface g0/0

ip address 192.168.20.1 255.255.255.0

no shutdown

interface s0/0/0

ip address 10.1.1.2 255.255.255.252

encapsulation ppp

no shutdown

end

Part 3: Configure CHAP Authentication

Step 1: Set Router Hostnames

R1:

enable

configure terminal

hostname R1

R2:

enable

configure terminal

hostname R2

Step 2: Configure CHAP Usernames

On R1:

```
username R2 password cisco
```

On R2:

```
username R1 password cisco
```

Step 3: Enable CHAP

On R1:

```
interface s0/3/0  
ppp authentication chap
```

On R2:

```
interface s0/3/0  
ppp authentication chap
```

Why CHAP?

CHAP provides authentication between routers before communication begins.

Benefits:

- ✓ More secure than PAP
 - ✓ Password is never sent in plain text
 - ✓ Prevents unauthorized routers from joining the WAN
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Part 4: Configure Static Routing

Without routes, each router only knows its directly connected networks.

Configure Static Route on R1

```
ip route 192.168.20.0 255.255.255.0 10.1.1.2
```

Meaning:

- Destination Network = 192.168.20.0/24
 - Next-Hop Router = R2 (10.1.1.2)
-

Configure Static Route on R2

```
ip route 192.168.10.0 255.255.255.0 10.1.1.1
```

Meaning:

- Destination Network = 192.168.10.0/24
 - Next-Hop Router = R1 (10.1.1.1)
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Part 5: Verify PPP Status

Check Interface Status

On either router:

```
show ip interface brief
```

Expected:

```
Serial0/3/0 up up
```

Verify PPP Authentication

```
show interfaces serial 0/3/0
```

Expected Output:

```
Encapsulation PPP
```

View Routing Table

```
show ip route
```

Expected:

```
C 192.168.10.0 is directly connected
```

```
S 192.168.20.0 via 10.1.1.2
```

and

```
C 192.168.20.0 is directly connected
```

```
S 192.168.10.0 via 10.1.1.1
```

Part 6: Test Connectivity

Step 1: Ping Between Routers

From R1:

ping 10.1.1.2

Expected:

!!!!

Success Rate = 100%

Step 2: Ping from PC-A to PC-B

On PC-A:

ping 192.168.20.10

Expected:

Reply from 192.168.20.10

4 successful replies.

Step 3: Ping from PC-B to PC-A

On PC-B:

ping 192.168.10.10

Expected:

Reply from 192.168.10.10

4 successful replies.

Step 4: Run Traceroute

On PC-A:

tracert 192.168.20.10

Expected Path:

192.168.10.1
10.1.1.2
192.168.20.10

This confirms traffic crosses the WAN serial link.

Troubleshooting Tips

If the serial link stays down:

✓ Verify both routers use PPP

show interfaces serial 0/0/0

✓ Verify CHAP usernames match router hostnames

show running-config

✓ Verify passwords match exactly

✓ Verify clock rate is configured on the DCE side

show controllers serial 0/0/0

✓ Verify static routes are correct

show ip route

✓ Confirm all interfaces show **up/up**

What You Learned

- How WAN serial connections work
- How PPP differs from HDLC
- How CHAP authenticates routers
- How point-to-point serial links are addressed
- How static routing works across WAN links
- How to troubleshoot PPP authentication issues
- How enterprise branch offices communicate over leased lines

Challenge Extensions (Optional)

Challenge 1: Break CHAP Authentication

On R2:

username R1 password wrongpassword

Questions:

- Does the serial link still come up?
- What errors appear?

Fix the password and verify connectivity.

Challenge 2: Add a Third Router

Create:

PC-A

|

R1

|

R2

|

R3

|

PC-C

Configure:

- PPP between R1-R2
- PPP between R2-R3
- Static routes on all routers

Test end-to-end connectivity.

Challenge 3: Use PAP Instead of CHAP

Remove CHAP and configure:

ppp authentication pap

Compare:

- PAP
- CHAP

Which is more secure and why?

Challenge 4: Observe PPP Packets in Simulation Mode

Switch Packet Tracer to **Simulation Mode**.

Generate pings between PCs and observe:

- PPP frames
- CHAP authentication process
- ICMP packets
- Routing decisions